

MA 403 Real Analysis

Homework 12

1. Find the radius of convergence of each of the following power series:

(a) $\sum n^3 x^n$ (b) $\sum \frac{2^n}{n!} x^n$

2. Given that the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence 2, find the radius of convergence of each of the following power series:

(a) $\sum_{n=0}^{\infty} a_n^k x^n$, where k is a fixed positive integer. (b) $\sum_{n=0}^{\infty} a_n x^{n^2}$.

3. Prove that every uniformly convergent sequence of bounded functions $\{f_n\}$ is uniformly bounded. That is, show that there is an $M > 0$ such that $|f_n(x)| \leq M$ for all x in the common domain S of the functions f_n and for all $n \in \mathbb{Z}_+$.
4. Construct a sequence $\{f_n\}$ of functions on $\mathbb{R}$ which converge pointwise to 0, but such that none of the functions f_n is bounded.
5. If $\{f_n\}$ and $\{g_n\}$ converge uniformly on a set E , prove that $\{f_n + g_n\}$ converges uniformly on E . If, in addition, $\{f_n\}$ and $\{g_n\}$ are sequences of bounded functions, prove that $\{f_n g_n\}$ converges uniformly on E .
6. Construct sequences $\{f_n\}, \{g_n\}$ which converge uniformly on some set E , such that $\{f_n g_n\}$ converges pointwise on E , but such that $\{f_n g_n\}$ does not converge uniformly on E .
7. (a) Let $f_n(x) = \frac{1}{nx+1}$ for $x \in (0, 1)$ and $n = 1, 2, \dots$. Prove that $\{f_n\}$ converges pointwise but not uniformly on $(0, 1)$.
- (b) Let $g_n(x) = \frac{x}{nx+1}$ for $x \in (0, 1)$ and $n = 1, 2, \dots$. Prove that g_n converges to 0 uniformly on $(0, 1)$.